



Combining Like Terms

Course: Intermediate Algebra · **Unit 1:** Expressions · **Deck:** U1L10_Combining_Like_Terms.pptx (22 slides)

 You do not need to know algebra to teach this lesson. Read what is written. Follow the slides. Check answers against the key. The learning expectations for today are the same as any other day — this plan gives you the tools to keep the lesson moving. The art of teaching stays with your teacher; the tools are right here.

QUICK START — BEFORE THE BELL (2 MINUTES)

- ☐ Open the slide deck: **U1L10_Combining_Like_Terms.pptx**
- ☐ Write the Bell Ringer on the board (see Slide 1 / Answer Key below)
- ☐ Test the projector
- ☐ Write your name on the board
- ☐ Find the answer key — it is printed inside this plan

MATERIALS FOR THIS LESSON

- ☐ Whiteboards, markers, erasers (1 set per student)
- ☐ Direct Instruction slide deck (Navy 1C2950 headings, Arial body)
- ☐ Tiered Independent Worksheet (T1/T2/T3 versions)
- ☐ Practice materials for: Small Group Practice
- ☐ Exit Ticket slips / LMS exit ticket
- ☐ Visual vocabulary anchor chart
- ☐ Theme visual: Itasca State Park

If anything is missing, check the teacher's desk or ask the front office.

PACING AT A GLANCE

Slide	By minute	Phase
1	0–4	Bell Ringer
7	15–16	End of Direct Instruction
19	42	End of Practice (worksheet + game/activity)
22	50	Exit Ticket & Wrap-Up

Pacing tip: bell ringer ~4 min · title/I-can/vocab ~4 min · collab ~4 min · worked example + guided practice ~7 min · culture + career ~6 min · GO/SET/EXTRA ~12 min · game/activity ~4 min · individual solve ~4 min · summary + exit ~4 min. Keep the timer visible; the deck drives the pace.

SUBSTITUTE SCRIPT (SLIDE BY SLIDE)

Slide 1 — Bell Ringer

On screen: WHITEBOARDS READY! — Tiered prompts (T1 one-step, T2 two-step, T3 challenge) shown on the slide.

Say: “Whiteboards ready — three tiers, four minutes. Choose your tier, work silently, show your steps, and hold your board up when the timer hits zero.”

Do: Walk the room as they work; cue “3... 2... 1 — boards up” and glance for accuracy.

Answer key: Answers are revealed on the slide after the timer. Bridge to today’s topic: Combining Like Terms.

Slide 2 — Title

On screen: INTERMEDIATE ALGEBRA — UNIT 1: LESSON 10 — COMBINING LIKE TERMS. Minnesota standards: 9.3.5.7. MERIT EDUCATION GROUP.

Say: “Welcome. Today’s lesson is Unit 1, Lesson 10: Combining Like Terms. Everything you need is on these slides — follow along and write when I ask.”

Do: Read the lesson title aloud. Pause for students to find a pencil.

Slide 3 — I Can Statements

On screen: I CAN STATEMENTS — Combine like terms; Identify like terms in an expression; Simplify expressions to their shortest form.

Say: “Here’s what we’ll be able to do by the end of the lesson. Read them silently; we’ll come back to these at the end.”

Do: Give 15 seconds, then continue.

Slide 4 — Vocabulary

On screen: VOCABULARY — Like terms — terms with the same variable(s) and exponents; Combine — add or subtract the coefficients; Simplify — write the expression in shortest form.

Say: “Quick vocabulary check. These are the words for today — I’ll read them, you repeat. [Read each term aloud.] We’ll use them all lesson.”

Do: Point to each term as you read it.

Slide 5 — Collaboration: Think-Pair-Share

On screen: ENGAGEMENT: THINK-PAIR-SHARE — prompt on the slide, tied to Combining Like Terms.

Say: “Think-Pair-Share. [Read the prompt aloud.] THINK silently for 30 seconds, then work with your partner or group, then we’ll share out.”

Do: Time 30s think, 45s pair, then take 2–3 volunteers.

Slide 6 — Worked Example (I Do)

On screen: WORKED EXAMPLE (I DO) — Step-by-step: Simplify $3x + 5 + 2x - 3 \rightarrow 5x + 2$.

Say: “A worked example — watch how it’s done step by step. The slide shows every step; I’ll read them as they appear.”

Do: Read each step aloud as it is revealed. Answer questions by pointing back to the slide.

Slide 7 — Guided Practice (We Do)

On screen: GUIDED PRACTICE (WE DO) — Together: Simplify $7a + 4 - 2a + 6$. Simplify $4y^2 + 3y - 2y^2 + y \rightarrow 5a + 10$ and $2y^2 + 4y$.

Say: “We do this one together. I’ll ask, you answer — then we check the slide.”

Do: Work step by step; call on students for the next step before revealing it.

Slide 8 — Culture

On screen: CULTURE: Liberian (Liberia). Liberian Americans are a growing Minnesota community, many settling in Brooklyn Park and Brooklyn Center.

Say: “A cultural connection. [Read the caption aloud.] Notice how math shows up in everyday life and community traditions.”

Do: If the slide shows an image only, describe what students see; no math required.

Slide 9 — Culture Word Problem

On screen: CULTURE WORD PROBLEM — A Liberian market sells rice in $4x$ -pound and $6x$ -pound bags. Simplify the total: $4x + 6x$.

Say: “Here’s the cultural word problem. Read it to yourself — we’ll solve it together. Use the reading strategy on the slide: circle, underline, box, write.”

Do: Give 30 seconds to read; walk through the strategy, then the solution.

Slide 10 — Career

On screen: CAREER: ASSEMBLY LINE TECHNICIAN — The Toro Company, Bloomington, MN. Entry wage: \$42,682/year (MN DEED OES, 25th percentile). Median wage: \$46,696/year.

Say: “A real career uses this math every day. [Read the career spotlight aloud.] The math we’re learning today is the math people use at work.”

Do: Read confidently; the salary and role are on the slide.

Slide 11 — Career Word Problem

On screen: CAREER WORD PROBLEM — A clinical trial coordinator at CVRx enrolls x patients in one group and y in another. Write an expression for total patients and evaluate when $x = 8$ and $y = 6$.

Say: “Here’s the career problem. Read it to yourself — we’ll solve it together. Use the reading strategy on the slide.”

Do: Give 30 seconds; walk the strategy; then the solution.

Answer key: Answer is revealed on the slide (solution mirrors today’s practice: $5x + 2$).

Slide 12 — GO Practice (Your Turn)

On screen: GO PRACTICE: YOUR TURN — Simplify $3x + 5 + 2x - 3$

Say: “Your turn — a problem just like the one we did. Work it on your whiteboard (or paper). Two minutes.”

Do: Circulate. Cue boards up at 2 minutes.

Answer key: $5x + 2$

Slide 13 — GO Answer

On screen: GO ANSWER — reveal: $5x + 2$.

Say: “Let’s check it. The slide walks through the solution step by step — I’ll read each step. Check your work against it; fix anything you missed.”

Do: Read the solution steps aloud.

Slide 14 — SET Practice

On screen: SET PRACTICE — two problems: Simplify $7a + 4 - 2a + 6$. Simplify $4y$ squared $+ 3y - 2y$ squared $+ y$.

Say: “Two practice problems to deepen your understanding. Work them on paper or whiteboard. You have 5 minutes; I’ll be walking around.”

Do: Circulate and support. Cue time at 3 and 1 minutes.

Slide 15 — SET Answer

On screen: SET ANSWER — reveal: $5a + 10$ and $2y$ squared $+ 4y$.

Say: “Let’s check. Read the steps on the slide; self-correct with a pen.”

Do: Read the steps aloud; answer questions by re-reading the relevant step.

Slide 16 — EXTRA Practice Challenge

On screen: EXTRA PRACTICE CHALLENGE — Simplify $5m + 3n - 2m + 7n - 4$

Say: “Challenge problem — for anyone who wants a stretch. Try it on your own paper. We’ll go over it in a moment.”

Do: Encourage all to attempt; it is optional.

Slide 17 — EXTRA Answer

On screen: EXTRA ANSWER — reveal: $3m + 10n - 4$.

Say: “Let’s check the challenge. [Read the steps.] Great work.”

Do: Read the solution steps aloud; celebrate attempts.

Slide 18 — Small Group Practice

On screen: SMALL GROUP PRACTICE — directions on the slide.

Say: “Small Group Practice. [Read the directions aloud.] We’ll check together at the end.”

Do: Split the class into small groups of 3–4. Each group works the practice set on the slide together. Circulate; check in with every group at least once. Pull the class back for a quick share-out of answers.

Slide 19 — Individual Solve: Tiered Worksheet

On screen: INDIVIDUAL SOLVE: TIERED WORKSHEET — TIER 1 Build Confidence (6 questions), TIER 2 Apply Your Skills (6 questions), TIER 3 Challenge Yourself (6 questions). PROBLEMS: TIERED WORKSHEET — COMBINING LIKE TERMS + IN YOUR LMS. SHOW YOUR WORK.

Say: “Independent work now — the tiered worksheet. Pick your tier, show your work, and submit in the LMS when you finish.”

Do: Circulate; support tier choices; collect or confirm LMS submission.

Slide 20 — Summary

On screen: SUMMARY — RECAP: key ideas from Combining Like Terms. REVIEW: the skills practiced today. [Slide lists the takeaways.]

Say: “Three big takeaways from today. [Read them aloud.] This is what you should remember tomorrow.”

Do: Read each takeaway; pause between them.

IF A STUDENT ASKS...

“Why are we doing this?” — “This is today’s learning expectation for Combining Like Terms. It builds toward what’s next in the unit, and your teacher will pick up right here tomorrow.”

Redirects (from the lesson plan):

Misconception	Redirect Question	Corrective Strategy
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| Combines unlike terms | “Do these two terms have the same variable and exponent?” | Color-code terms by variable. |

| Changes the variable when combining | “What stays the same when we combine like terms?” | The variable part copies over; only coefficients add. |

| Drops the sign in front of a term | “What sign does the term carry?” | Rewrite the expression with every sign visible. |

IF A STUDENT FINISHES EARLY

- Students who finish early do the **Challenge** side of the worksheet — do not invent extra work.

- If the whole class is ahead: re-read the Summary slide and ask students to explain one takeaway in their own words.
- **Do not** move ahead in the curriculum.

IF THE PROJECTOR DIES

Write the Bell Ringer problems on the board (from the Answer Key).

Read the worksheet directions aloud; students work in pairs.

Use the printed answer key to check as a class.

The lesson still runs — the script lives in this plan, not the projector.

WITH A PARAPROFESSIONAL OR CO-TEACHER

If an adult is in the room supporting students, they know the students best — **follow their lead** on individual support. Check the roster for accommodations; extended time on the Exit Ticket is fine if indicated.

SUB REPORT (LEAVE FOR THE TEACHER)

What did we complete? (bell ringer / DI / practice / game / exit ticket)

Who struggled or needed extra support?

Behavior notes (positive first, then concerns)

Anything tech-related that went wrong

TEACHER'S OWN NOTES

(The regular teacher may add specifics here — their timer, signals, roster quirks.)

Generated by the MEG Substitute Plan pipeline from the Lesson Plan + Teacher Edition — synced to the published slide decks (2026-08-19).